r 2016 (New)-e.pdf

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
10.	*(c)*			M_r HF is 20.01 Solution contains 500g HF in 1000g solution (using V = m/D) 1000 g of the solution has a volume of 855 cm^3 or Number of moles of HF in 1000g / 855 cm^3 solution is $\frac{500}{20.01}$ = 24.98 (1) 855 cm^3 contain 24.98 mol ∴ Concentration of HF = 29.2 mol dm^{-3} (1) ecf possible		2		2	2	
	(d)	(i)		(There are 6 bonding pairs of electrons and no lone pairs –) position of minimum repulsion taken up (1) Drawing shows clear octahedral shape (1) Bond angle is 90° equatorial / equatorial or 90° equatorial / vertical (accept 180° if vertical bonds only considered) (1)	1	2		3		